

RESEARCH PoC

ResolveFlow AI

Multi-Agent RAG Pipeline for C2C Marketplace Support Triage

Pinecone

Gemini 2.5

LangGraph

Streamlit

Nelson Chou

The Problem

Customer support triage relies on inconsistent human judgement at scale



Inconsistent Decisions

Escalation choices vary by agent — same issue handled differently depending on who picks it up



Manual Supervisor Work

Supervisor resolution follows replicable logic yet still requires significant human time and coordination



No Institutional Memory

Past cases and resolutions aren't searchable — every issue starts from zero with no historical context

Result: inconsistent customer experience at scale

Solution Overview

An agentic AI pipeline that standardizes escalation decisions and automates supervisor-level resolution



⚡ Severity < 4: Early Exit after Agent 3 — skips Supervisor, ~40% faster ·



Per-agent latency + token usage tracked in real time.



Trust & Safety flag → Human Review circuit breaker

① Customer-facing answer

② Internal action plan

③ Audit ticket → Pinecone

Built for C2C Marketplace Reality

Generic support tools don't understand buyer · seller · platform dynamics

What makes this different

- ✓ 3-party context: buyer / seller / platform each have different escalation logic
- ✓ Transaction value as severity signal — high-value disputes auto-escalate
- ✓ Seller rating threshold triggers Trust & Safety flag automatically
- ✓ Repeat dispute detection raises severity score for known offenders
- ✓ Institutional memory: similar past cases retrieved and factored in

Demo scenario

"I paid in-app but seller disappeared after receiving payment"

buyer · \$250 · seller rating 2.8 · 3 past disputes

Pipeline output

- 🔴 Severity 5 · Trust & Safety flagged
- 🛡️ Routed to trust_safety_team
- 💰 Full refund recommended
- 📄 Case logged to institutional memory

Key Design Decisions & Trade-offs

Decision	Option Chosen	Trade-off
What to embed?	Questions, not answers	Misses answer-content matches
Severity scoring	LLM-based (Gemini)	Non-deterministic across runs
Past case retrieval	Configurable similarity threshold (default: 0.81)	Higher threshold = fewer but more relevant past cases retrieved
Low-severity routing	Early exit after Agent 3	Saves ~40% latency for sev 1–2

Decision in Practice

Threshold 0.75 caused wrong severity scores



Raised threshold to 0.81



Unrelated past cases no longer affect evaluation ✓

Severity 1–2 queries ran all 5 agents unnecessarily



Early exit after Agent 3



~40% latency reduction for routine queries ✓

Evaluation Framework & Results

20 labeled test cases · all 5 severity levels · boundary cases + conflicting signals

0.25

Severity MAE

target < 0.5

89%

Escalation Precision

1 over-escalation

100%

Escalation Recall

zero missed

Honest Limitations

- Test cases are deliberately distinct — real-world boundary cases would likely reduce accuracy
- Severity scoring is LLM-based and non-deterministic — same query can score differently across runs
- Next step: RAGAS framework to evaluate retrieval precision and answer faithfulness at scale

Tech Stack & Production Roadmap

Pinecone

Vector DB · 2 namespaces

Gemini 2.5 Flash

LLM + Embeddings

LangGraph

Agent orchestration

Streamlit

Demo UI
Latency & Token Monitoring

P0 — Must Have

- PII Redaction before embedding
- Per-agent error handling + fallback ← already implemented 
- Per-agent latency + token tracking ← already implemented 
- FastAPI REST endpoint

P1 — Production

- Zendesk / ticketing integration
- Human feedback loop on prompts
- Monitoring: latency + drift

P2 — Scale

- Multilingual embedding support
- Hybrid search: vector + keyword
- Fine-tuned severity classifier

ResolveFlow AI

From PoC to Handoff — a blueprint for AI-powered support triage
in C2C marketplace platforms



[Notebook on GitHub](#)



[Live Demo on Streamlit](#)



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